

Vectors in n Dimensions

ANSWER KEY



Vector addition and scalar multiplication

- 1 Let $\mathbf{u} = \langle 2, -1, 3 \rangle$, $\mathbf{v} = \langle 1, 4, -2 \rangle$. Find $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$.
 $\mathbf{u} + \mathbf{v} = \langle 3, 3, 1 \rangle$. $3\mathbf{u} = \langle 6, -3, 9 \rangle$, $2\mathbf{v} = \langle 2, 8, -4 \rangle$, so $3\mathbf{u} - 2\mathbf{v} = \langle 4, -11, 13 \rangle$.
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Dot product and norm

- 2 For $\mathbf{u} = \langle 1, 2, 2 \rangle$, $\mathbf{v} = \langle 3, -1, 0 \rangle$, find $\mathbf{u} \cdot \mathbf{v}$ and $|\mathbf{u}|$.
 $\mathbf{u} \cdot \mathbf{v} = 1(3) + 2(-1) + 2(0) = 1$. $|\mathbf{u}| = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$.
- 3 Find the angle between $\mathbf{u} = \langle 1, 0, 0 \rangle$ and $\mathbf{v} = \langle 1, 1, 0 \rangle$ using $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.
 $\mathbf{u} \cdot \mathbf{v} = 1$, $|\mathbf{u}| = 1$, $|\mathbf{v}| = \sqrt{2}$. $\cos \theta = \frac{1}{\sqrt{2}}$, so $\theta = 45^\circ (\pi/4)$.
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The Cauchy–Schwarz Inequality

- 4 State the Cauchy–Schwarz Inequality.
For all vectors $\mathbf{u}$, $\mathbf{v}$: $|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|$.
- 5 Verify the Cauchy–Schwarz Inequality for $\mathbf{u} = \langle 3, 4 \rangle$, $\mathbf{v} = \langle 1, 2 \rangle$.
 $\mathbf{u} \cdot \mathbf{v} = 3 + 8 = 11$. $|\mathbf{u}| = 5$, $|\mathbf{v}| = \sqrt{5}$, so $|\mathbf{u}||\mathbf{v}| = 5\sqrt{5} \approx 11.18$. Since $11 \leq 11.18$, the inequality holds (it is strict here because $\mathbf{u}$ and $\mathbf{v}$ are not parallel).
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Unit vectors

- 6 Find the unit vector in the direction of $\mathbf{v} = \langle 3, -4 \rangle$.
 $|\mathbf{v}| = \sqrt{9 + 16} = 5$. Unit vector: $\left\langle \frac{3}{5}, -\frac{4}{5} \right\rangle$.
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Orthogonality

- 7 Determine whether $\mathbf{u} = \langle 2, -3, 1 \rangle$ and $\mathbf{v} = \langle 1, 2, 4 \rangle$ are orthogonal.
 $\mathbf{u} \cdot \mathbf{v} = 2(1) + (-3)(2) + 1(4) = 2 - 6 + 4 = 0$. Since the dot product is 0, they are orthogonal.
- 8 For $\mathbf{u} = \langle 1, -1, 2 \rangle$, find $|\mathbf{u}|$ and the unit vector $\hat{\mathbf{u}}$.
 $|\mathbf{u}| = \sqrt{1 + 1 + 4} = \sqrt{6}$. $\hat{\mathbf{u}} = \left\langle \frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right\rangle$.
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More vector arithmetic

- 9 Let $\mathbf{u} = \langle 0, 3, -1 \rangle$, $\mathbf{v} = \langle 4, -2, 5 \rangle$. Find $2\mathbf{u} + 3\mathbf{v}$ and $\mathbf{u} - \mathbf{v}$.
 $2\mathbf{u} = \langle 0, 6, -2 \rangle$, $3\mathbf{v} = \langle 12, -6, 15 \rangle$; $2\mathbf{u} + 3\mathbf{v} = \langle 12, 0, 13 \rangle$. $\mathbf{u} - \mathbf{v} = \langle -4, 5, -6 \rangle$.
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More dot product and norm practice

- 10 For $\mathbf{u} = \langle 2, 2, 1 \rangle$, $\mathbf{v} = \langle 0, 3, 4 \rangle$, find $\mathbf{u} \cdot \mathbf{v}$, $|\mathbf{u}|$, $|\mathbf{v}|$.
 $\mathbf{u} \cdot \mathbf{v} = 0 + 6 + 4 = 10$. $|\mathbf{u}| = \sqrt{4 + 4 + 1} = \sqrt{9} = 3$. $|\mathbf{v}| = \sqrt{0 + 9 + 16} = \sqrt{25} = 5$.
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More angle-between-vectors practice

- 11 Find the angle between $\mathbf{u} = \langle 1, 1, 0 \rangle$ and $\mathbf{v} = \langle 0, 1, 1 \rangle$.
 $\mathbf{u} \cdot \mathbf{v} = 0 + 1 + 0 = 1$. $|\mathbf{u}| = \sqrt{2}$, $|\mathbf{v}| = \sqrt{2}$. $\cos \theta = \frac{1}{2}$, so $\theta = 60^\circ (\pi/3)$.
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The triangle inequality

- 12 State the triangle inequality for vectors.
 $|\mathbf{u} + \mathbf{v}| \leq |\mathbf{u}| + |\mathbf{v}|$.
- 13 Verify the triangle inequality for $\mathbf{u} = \langle 3, 0 \rangle$, $\mathbf{v} = \langle 0, 4 \rangle$.
 $\mathbf{u} + \mathbf{v} = \langle 3, 4 \rangle$, $|\mathbf{u} + \mathbf{v}| = 5$. $|\mathbf{u}| + |\mathbf{v}| = 3 + 4 = 7$. Since $5 \leq 7$, the inequality holds.
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Verifying a unit vector

- 14 For $\mathbf{u} = \langle 1, 2, -2 \rangle$, find $|\mathbf{u}|$, and verify that $\hat{\mathbf{u}} = \mathbf{u}/|\mathbf{u}|$ satisfies $|\hat{\mathbf{u}}| = 1$.
 $|\mathbf{u}| = \sqrt{1 + 4 + 4} = 3$. $\hat{\mathbf{u}} = \left\langle \frac{1}{3}, \frac{2}{3}, -\frac{2}{3} \right\rangle$. $|\hat{\mathbf{u}}| = \sqrt{\frac{1}{9} + \frac{4}{9} + \frac{4}{9}} = \sqrt{\frac{9}{9}} = 1$. ✓
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Vector projection

- 15 Find the projection of $\mathbf{u} = \langle 3, 4 \rangle$ onto $\mathbf{v} = \langle 1, 0 \rangle$, using $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \mathbf{v}$.

$$\mathbf{u} \cdot \mathbf{v} = 3, |\mathbf{v}|^2 = 1. \text{proj}_{\mathbf{v}} \mathbf{u} = \frac{3}{1} \langle 1, 0 \rangle = \langle 3, 0 \rangle.$$

- 16 Find the projection of $\mathbf{u} = \langle 2, 3 \rangle$ onto $\mathbf{v} = \langle 1, 1 \rangle$.

$$\mathbf{u} \cdot \mathbf{v} = 2 + 3 = 5, |\mathbf{v}|^2 = 2. \text{proj}_{\mathbf{v}} \mathbf{u} = \frac{5}{2} \langle 1, 1 \rangle = \left\langle \frac{5}{2}, \frac{5}{2} \right\rangle.$$

Distance between two points as a vector norm

- 17 Find the distance between $P(1, 2, 3)$ and $Q(4, -1, 5)$ using $|\overrightarrow{PQ}|$.

$$\overrightarrow{PQ} = \langle 3, -3, 2 \rangle. |\overrightarrow{PQ}| = \sqrt{9 + 9 + 4} = \sqrt{22}.$$

Linear combinations of vectors

- 18 Write $\mathbf{w} = \langle 7, 4 \rangle$ as a linear combination $a\mathbf{u} + b\mathbf{v}$ of $\mathbf{u} = \langle 1, 0 \rangle$ and $\mathbf{v} = \langle 0, 1 \rangle$.
 $\mathbf{w} = 7\mathbf{u} + 4\mathbf{v}$, i.e. $a = 7$, $b = 4$ (the standard basis vectors, so the coefficients are just $\mathbf{w}$'s own components).
- 19 Write $\mathbf{w} = \langle 5, 1 \rangle$ as a linear combination $a\mathbf{u} + b\mathbf{v}$ of $\mathbf{u} = \langle 1, 1 \rangle$ and $\mathbf{v} = \langle 1, -1 \rangle$.
Need $a + b = 5$ and $a - b = 1$. Adding: $2a = 6 \Rightarrow a = 3$; then $b = 2$. So $\mathbf{w} = 3\mathbf{u} + 2\mathbf{v}$. Check:
 $3\langle 1, 1 \rangle + 2\langle 1, -1 \rangle = \langle 3, 3 \rangle + \langle 2, -2 \rangle = \langle 5, 1 \rangle$. ✓
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Vectors from two points

- 20 Find the vector $\overrightarrow{PQ}$ from $P(2, -1, 0)$ to $Q(5, 3, -2)$.

$$\overrightarrow{PQ} = \langle 5 - 2, 3 - (-1), -2 - 0 \rangle = \langle 3, 4, -2 \rangle.$$

- 21 Find $|\overrightarrow{PQ}|$ for the vector found above.

$$|\overrightarrow{PQ}| = \sqrt{9 + 16 + 4} = \sqrt{29}.$$

Scalar multiples and direction

- 22 For $\mathbf{v} = \langle 2, -3 \rangle$, is $-4\mathbf{v}$ in the same direction as $\mathbf{v}$, or the opposite?
Since $-4 < 0$, $-4\mathbf{v} = \langle -8, 12 \rangle$ points in the opposite direction to $\mathbf{v}$ (and is 4 times as long).
- 23 Are $\mathbf{u} = \langle 2, -6 \rangle$ and $\mathbf{v} = \langle -1, 3 \rangle$ parallel?
 $\mathbf{u} = -2\mathbf{v}$ (check: $-2\langle -1, 3 \rangle = \langle 2, -6 \rangle$ ✓), so yes — $\mathbf{u}$ and $\mathbf{v}$ are parallel (scalar multiples of each other).
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The midpoint of a segment as a vector average

- 24 Find the midpoint M of the segment from $P(1, -2, 3)$ to $Q(5, 4, -1)$, using $M = \frac{1}{2}(P + Q)$ (averaging componentwise).

$$M = \frac{1}{2}\langle 1 + 5, -2 + 4, 3 + (-1) \rangle = \frac{1}{2}\langle 6, 2, 2 \rangle = \langle 3, 1, 1 \rangle.$$

Normalizing a vector to a given length

25 Find the vector of length 10 in the same direction as $\mathbf{v} = \langle 3, 4 \rangle$.

$|\mathbf{v}| = 5$, so the unit vector is $\left\langle \frac{3}{5}, \frac{4}{5} \right\rangle$. Scaling to length 10: $10 \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle = \langle 6, 8 \rangle$.